1.

The research problem is that inter-industry labour flows were poorly understood, so it was unknown whether human capital has an industry-specific component constraining worker mobility. Earlier relatedness measures (co-exports, co-location, input-output) captured relatedness only indirectly, never through the skills workers actually share. Using ~20 million German workers, the paper shows that labour flows are highly structured. They are concentrated in narrow industry pairs (5% of employment absorbs 62% of switchers), stable over time, identical across occupations and wage levels, and independent of co-location, indicating a genuine industry-specific component of human capital. Crucially, this labour-flow-based skill relatedness predicts regional industry entry and growth better than co-location or input-output measures.

2.

Janssen, M. J. J., & Abbasiharofteh, M. (2022). Boundary spanning R&D collaboration: Key enabling technologies and missions as alleviators of proximity effects? *Technological Forecasting & Social Change, 180*, Article 121689. https://doi.org/10.1016/j.techfore.2022.121689

They extended skill relatedness to a new context and outcome: using Dutch labour-mobility data as a measure of cognitive proximity in gravity models of R&D tie formation, they find that key enabling technologies (Digital, Chemical, Advanced Materials) weaken the cognitive-proximity constraint on partner choice, while mission-labelled projects actually unite cognitively similar firms.

Hane-Weijman, E., Eriksson, R. H., & Rigby, D. (2022). How do occupational relatedness and complexity condition employment dynamics in periods of growth and recession? *Regional Studies, 56*(7), 1176–1189. https://doi.org/10.1080/00343404.2021.1984420

They applied the skill-relatedness method to a new unit of analysis: occupations (112 Swedish 3-digit) instead of industries. They combined it with occupational complexity in a Smart Specialisation framework: relatedness drove employment growth mainly pre-2008 via exits from unrelated occupations, while after the financial crisis, entering more complex occupations became positively associated with growth.

3.

Chakraborty, A., Inoue, H., & Fujiwara, Y. (2020). Economic complexity of prefectures in Japan. *PLOS ONE, 15*(8), Article e0238017. https://doi.org/10.1371/journal.pone.0238017

Gap statement: Previous work has shown that labour flows yield stable skill relatedness matrices that predict regional industry entry and growth better than co-location or input-output measures (Neffke et al., 2017), and that prefectural complexity in Japan predicts income and growth potential (Chakraborty et al., 2020). But we do not know whether Japanese prefectures diversify into sectors that are skill related to their existing industrial base, no labour-flow-based skill relatedness matrix exists for Japan, where prefectural complexity has only been inferred from firms' sectoral composition, never from the workers who carry those capabilities. This matters because retraining and cluster policies for prefectures below the ECI-income curve (e.g., Osaka, Fukuoka) are only effective if they target sectors local workers can realistically move into.

Research question: Does labour-flow-based skill relatedness predict which new industries Japanese prefectures enter and grow, better than co-location or input-output relatedness?

4.

Data source: Japan's matched employer-employee records from the Employees' Pension and Employment Insurance system, Japan's closest analogue to the German IAB data used by Neffke et al. (2017). It tracks workers across employers, recording the location and 4-digit industry (JSIC) of both origin and destination establishments, allowing prefecture × industry labour-flow matrices to be.

Who would use the answer: Regional and national policy makers: prefectural governments (e.g., Osaka, Fukuoka) choosing which industries to attract in smart specialisation strategies, and agencies designing retraining programs that target industries local workers can realistically move into.